

FLORIDA ENERGY EFFICIENCY CODE FOR BUILDING CONSTRUCTION

Florida Department of Business and Professional Regulation - Residential Performance Method

Project Name: 3316 W Nassau Street 3 Ton 15 2 Street: 3316 W Nassau Street City, State, Zip: Tampa, FL, 33607 Owner: Design Location: FL, Tampa	Builder Name: Permit Office: Permit Number: Jurisdiction: County: Hillsborough(Florida Climate Zone 2)
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1. New construction or existing Existing (Projected) 2. Single family or multiple family Detached 3. Number of units, if multiple family 1 4. Number of Bedrooms 4 5. Is this a worst case? No 6. Conditioned floor area above grade (ft²) 1634 Conditioned floor area below grade (ft²) 0 7. Windows(136.2 sqft.) Description Area a. U-Factor: Dbl, U=0.34 136.18 ft² SHGC: SHGC=0.23 b. U-Factor: N/A ft² SHGC: c. U-Factor: N/A ft² SHGC: Area Weighted Average Overhang Depth: 1.453 ft Area Weighted Average SHGC: 0.230 8. Skylights Description Area U-Factor:(AVG) N/A N/A ft² SHGC(AVG): N/A 9. Floor Types Insulation Area a. Slab-On-Grade Edge Insulation R= 0.0 1634.00 ft² b. N/A R= ft² c. N/A R= ft²	10. Wall Types(1360.0 sqft.) Insulation Area a. Concrete Block - Int Insul, Exterior R=4.1 1360.00 ft² b. N/A c. N/A d. N/A 11. Ceiling Types(1634.0 sqft.) Insulation Area a. Roof Deck (Unvented) R=23.0 1144.00 ft² b. Single assembly, with (Unvented) R=0.0 490.00 ft² c. N/A 12. Roof(Comp. Shingles, Unvent) Deck R=23.0 1684 ft² 13. Ducts, location & insulation level R ft² a. Sup: Attic, Ret: Attic, AH: Main Pit 6 280 b. c. 14. Cooling Systems kBtu/hr Efficiency a. Central Unit 34.5 SEER2:15.20 15. Heating Systems kBtu/hr Efficiency a. Electric Heat Pump 34.0 HSPF2:7.80 16. Hot Water Systems a. Electric Cap: 40 gallons EF: 0.950 b. Conservation features None 17. Credits CF, Pstat
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Glass/Floor Area: 0.083	Total Proposed Modified Loads: 50.13	PASS
	Total Baseline Loads: 52.57	

NOTE: Proposed residence must have annual total normalized Modified Loads that are less than or equal to 95 percent of the annual total loads of the standard reference design in order to comply.

I hereby certify that the plans and specifications covered by this calculation are in compliance with the Florida Energy Code. PREPARED BY: <u>Navarrete Drafting Inc.</u> DATE: <u>9/18/25</u> <u>Gabriel Navarrete 813-549-9736</u> I hereby certify that this building, as designed, is in compliance with the Florida Energy Code. OWNER/AGENT: _____ DATE: _____	Review of the plans and specifications covered by this calculation indicates compliance with the Florida Energy Code. Before construction is completed this building will be inspected for compliance with Section 553.908 Florida Statutes. BUILDING OFFICIAL: _____ DATE: _____
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- Compliance requires certification by the air handler unit manufacturer that the air handler enclosure qualifies as certified factory-sealed in accordance with R403.3.2.1.
- 1 of the 1 duct systems requires a Duct Leakage Test Report. Systems with Default duct leakage do not require this report.
- Compliance requires an Air Barrier and Insulation Inspection Checklist in accordance with R402.4.1.1 and this project requires a PERFORMANCE envelope leakage test report with envelope leakage no greater than 5.00 ACH50 (R402.4.1.2).

INPUT SUMMARY CHECKLIST REPORT

PROJECT												
Title:	3316 W Nassau Street 3 Ton 15 2					Address type:		Street Address				
Building Type:	User		Bedrooms:	4		Lot #:		---				
Owner:			Conditioned Area:	1634		Block/SubDivision:		---				
Builder Home ID:			Total Stories:	1		PlatBook:		---				
Builder Name:			Worst Case:	No		Street:		3316 W Nassau Street				
Permit Office:			Rotate Angle:	0		County:		Hillsborough				
Jurisdiction:			Cross Ventilation:			City, State, Zip:		Tampa, FL, 33607				
Family Type:	Detached		Whole House Fan:									
New/Existing:	Existing (Projected)		Terrain:	Suburban								
Year Construct:	2025		Shielding:	Suburban								
Comment:												
CLIMATE												
✓ Design Location	Tmy Site		Design Temp		Int Design Temp		Heating		Design		Daily temp	
			97.5%	2.5%	Winter	Summer	Degree Days		Moisture		Range	
___ FL, Tampa	FL_TAMPA_INTERNATIONAL		39	91	70	75	645.5		54		Medium	
BLOCKS												
✓ Number	Name		Area		Volume							
___ 1	Block1		1634		13072 cu ft							
SPACES												
✓ Number	Name		Area		Volume		Kitchen		Occupants		Bedrooms	
___ 1	Main Pitched Roof		1144	9152	Yes	4	3	Yes	Yes	Yes	Yes	Yes
___ 2	Main Flat Roof		490	3920	No	2	1	Yes	Yes	Yes	Yes	Yes
FLOORS (Total Exposed Area = 1634 sq.ft.)												
✓ #	Floor Type		Space		Exposed Perim(ft)		Area		R-Value Perim. Joist		U-Factor	
___ 1	Slab-On-Grade Edge Ins		Main Pitched Roof	96	1144 sqft	0	---	0.651	0 (ft)/0 (ft)	0.50	0.50	0.00
___ 2	Slab-On-Grade Edge Ins		Main Flat Roof	74	490 sqft	0	---	0.710	0 (ft)/0 (ft)	1.00	0.00	0.00
ROOF												
✓ #	Type		Materials		Roof Area		Gable Area		Roof Color		Rad Barr	
___ 1	Gable or shed		Composition shingles	1684 ft²	204 ft²	Medium	N	0.75	No	0.9	No	23
												14.04
ATTIC												
✓ #	Type		Ventilation		Vent Ratio (1 in)		Area		RBS		IRCC	
___ 1	Full attic		Unvented	0	1634 ft²	N	N					

INPUT SUMMARY CHECKLIST REPORT

CEILING													(Total Exposed Area = 1634 sq.ft.)			
✓ #	Ceiling Type	Space	R-Value	Ins. Type	Area	U-Factor	Framing Frac.	Truss Type								
___ 1	Flat ceiling under attic(Unvented)	Main Pitched Roof	0.0	Blown	1144.0ft²	0.039	0.11	Wood								
___ 2	Single assembly, with airspace(Unvented)	Main Flat Roof	0.0	Blown	490.0ft²	0.039	0.11	Wood								

WALLS													(Total Exposed Area = 1360 sq.ft.)			
✓ #	Ornt	Adjacent To	Wall Type	Space	Cavity R-Value	Width Ft	In	Height Ft	In	Area sq.ft.	U-Factor	Sheath R-Value	Frm. Frac.	Solar Absor.	Below Grade	
___ 1	N	Exterior	Conc. Blk - Int Ins	Main Pitched Roof	4.1	18.0	0	8.0	0	144.0	0.149		0	0.35	0 %	
___ 2	N	Exterior	Conc. Blk - Int Ins	Main Pitched Roof	4.1	8.0	0	8.0	0	64.0	0.149		0	0.35	0 %	
___ 3	E	Exterior	Conc. Blk - Int Ins	Main Pitched Roof	4.1	44.0	0	8.0	0	352.0	0.149		0	0.35	0 %	
___ 4	S	Exterior	Conc. Blk - Int Ins	Main Pitched Roof	4.1	26.0	0	8.0	0	208.0	0.149		0	0.35	0 %	
___ 5	S	Exterior	Conc. Blk - Int Ins	Main Flat Roof	4.1	10.0	0	8.0	0	80.0	0.149		0	0.35	0 %	
___ 6	W	Exterior	Conc. Blk - Int Ins	Main Flat Roof	4.1	49.0	0	8.0	0	392.0	0.149		0	0.35	0 %	
___ 7	N	Exterior	Conc. Blk - Int Ins	Main Flat Roof	4.1	10.0	0	8.0	0	80.0	0.149		0	0.35	0 %	
___ 8	E	Exterior	Conc. Blk - Int Ins	Main Flat Roof	4.1	5.0	0	8.0	0	40.0	0.149		0	0.35	0 %	

DOORS													(Total Exposed Area = 60 sq.ft.)			
✓ #	Ornt	Adjacent To	Door Type	Space	Storms	U-Value	Width Ft	In	Height Ft	In	Area					
___ 1	N	Exterior	Insulated	Main Pitched Roof	None	0.40	3.00	0	6.00	8	20.0ft²					
___ 2	S	Exterior	Insulated	Main Pitched Roof	None	0.40	3.00	0	6.00	8	20.0ft²					
___ 3	S	Exterior	Insulated	Main Flat Roof	None	0.40	3.00	0	6.00	8	20.0ft²					

WINDOWS													(Total Exposed Area = 136 sq.ft.)				
✓ #	Ornt	Wall ID	Frame	Panes	NFRC U-Factor	SHGC	Imp	Storm	Total Area (ft²)	Same Units	Width (ft)	Height (ft)	--Overhang-- Depth (ft)	Sep. (ft)	Interior Shade	Screen	
___ 1	N	1	Vinyl	Low-E Double	Y	0.34	0.23	Y	N	9.3	1	3.00	3.08	6.3	0.3	Drapes/blinds	None
___ 2	N	2	Vinyl	Low-E Double	Y	0.34	0.23	Y	N	18.5	2	3.00	3.08	1.0	2.3	Drapes/blinds	None
___ 3	E	3	Vinyl	Low-E Double	Y	0.34	0.23	Y	N	37.0	4	3.00	3.08	1.3	0.7	Drapes/blinds	None
___ 4	S	4	Vinyl	Low-E Double	Y	0.34	0.23	Y	N	18.5	2	3.00	3.08	1.0	3.3	Drapes/blinds	None
___ 5	W	6	Vinyl	Low-E Double	Y	0.34	0.23	Y	N	27.8	3	3.00	3.08	1.0	1.0	Drapes/blinds	None
___ 6	W	6	Vinyl	Low-E Double	Y	0.34	0.23	Y	N	6.7	1	3.08	2.17	1.0	1.0	Drapes/blinds	None
___ 7	N	7	Vinyl	Low-E Double	Y	0.34	0.23	Y	N	18.5	2	3.00	3.08	1.0	1.0	Drapes/blinds	None

INFILTRATION										
✓ #	Scope	Method	SLA	CFM50	ELA	EqLA	ACH	ACH50	Space(s)	Infiltration Test Volume
___ 1	Wholehouse	Proposed ACH(50)	0.00025	1089	59.76	112.20	0.0954	5.0	All	13072 cu ft

MASS					
✓ #	Mass Type	Area	Thickness	Furniture Fraction	Space
___ 1	Default(8 lbs/sq.ft.)	0 ft²	0 ft	0.30	Main Pitched Roof
___ 2	Default(8 lbs/sq.ft.)	0 ft²	0 ft	0.30	Main Flat Roof

INPUT SUMMARY CHECKLIST REPORT

HEATING SYSTEM

✓ #	System Type	Subtype/Speed	AHRI #	Efficiency	Capacity kBtu/hr	---Geothermal HeatPump--- Entry Power Volt Current				Ducts	Block
___ 1	Electric Heat Pump	Split/Single		HSPF2: 7.80	34.0	0.00	0.00	0.00		sys#1	1

COOLING SYSTEM

✓ #	System Type	Subtype/Speed	AHRI #	Efficiency	Capacity kBtu/hr	Air Flow cfm	SHR	Duct	Block
___ 1	Central Unit	Split/Single		SEER2:15.2	34.5	1035	0.75	sys#1	1

HOT WATER SYSTEM

✓ #	System Type	Subtype	Location	EF(UEF)	Cap	Use	SetPnt	Fixture Flow	Pipe Ins.	Pipe length
___ 1	Electric	None	Main Flat Roof	0.95 (0.93)	40.00 gal	60 gal	120 deg	Standard	None	40
	Recirculation System	Recirc Control Type	Loop length	Branch length	Pump power	DWHR	Facilities Connected	Equal Flow	DWHR Eff	Other Credits
___ 1	No		NA	NA	NA	No	NA	NA	NA	None

DUCTS

✓ Duct #	Location	Supply----- R-Value Area	Return----- R-Value Area	Leakage Type	Air Handler	CFM 25 TOT	CFM 25 OUT	QN OUT	RLF	HVAC # Heat Cool
___ 1	Attic	6.0 280 ft²	Attic 6.0 70 ft²	Default Leakage	Main Pitch	(Default)	(Default)			1 1

TEMPERATURES

Programable Thermostat: Y					Ceiling Fans: Y									
Cooling	[] Jan	[] Feb	[] Mar	[] Apr	[] May	[X] Jun	[X] Jul	[X] Aug	[X] Sep	[] Oct	[] Nov	[] Dec		
Heating	[X] Jan	[X] Feb	[X] Mar	[] Apr	[] May	[] Jun	[] Jul	[] Aug	[] Sep	[] Oct	[X] Nov	[X] Dec		
Venting	[] Jan	[] Feb	[X] Mar	[X] Apr	[] May	[] Jun	[] Jul	[] Aug	[] Sep	[X] Oct	[X] Nov	[] Dec		
Thermostat Schedule: HERS 2006 Reference														
✓ Schedule Type		1	2	3	4	5	6	7	8	9	10	11	12	
___ Cooling (WD)	AM PM	78 80	78 80	78 78	78 78	78 78	78 78	78 78	78 78	80 78	80 78	80 78	80 78	
___ Cooling (WEH)	AM PM	78 78	78 78	78 78	78 78	78 78	78 78	78 78	78 78	78 78	78 78	78 78	78 78	
___ Heating (WD)	AM PM	66 68	66 68	66 68	66 68	66 68	68 68	68 68	68 68	68 68	68 68	68 66	68 66	
___ Heating (WEH)	AM PM	66 68	66 68	66 68	66 68	66 68	68 68	68 68	68 68	68 68	68 68	68 66	68 66	

ENERGY PERFORMANCE LEVEL (EPL) DISPLAY CARD

ESTIMATED ENERGY PERFORMANCE INDEX* = 95

The lower the EnergyPerformance Index, the more efficient the home.

3316 W Nassau Street,Tampa,FL,33607

1. New construction or existing	Existing (Projected)	10. Wall Types(1360.0 sqft.)	Insulation	Area
2. Single family or multiple family	Detached	a. Concrete Block - Int Insul, Exterior	R=4.1	1360.00 ft ²
3. Number of units, if multiple family	1	b. N/A		
4. Number of Bedrooms	4	c. N/A		
5. Is this a worst case?	No	d. N/A		
6. Conditioned floor area above grade (ft ²)	1634	11. Ceiling Types(1634.0 sqft.)	Insulation	Area
Conditioned floor area below grade (ft ²)	0	a. Roof Deck (Unvented)	R=23.0	1144.00 ft ²
7. Windows**	Description	b. Single assembly, with (Unvented)	R=0.0	490.00 ft ²
a. U-Factor:	Dbl, U=0.34	c. N/A		
SHGC:	SHGC=0.23	12. Roof(Comp. Shingles, Unvent) Deck	R=23.0	1684 ft ²
b. U-Factor:	N/A	13. Ducts, location & insulation level	R	ft ²
SHGC:		a. Sup: Attic, Ret: Attic, AH: Main Pitc	6	280
c. U-Factor:	N/A	b.		
SHGC:		c.		
Area Weighted Average Overhang Depth:	1.453 ft	14. Cooling Systems	kBtu/hr	Efficiency
Area Weighted Average SHGC:	0.230	a. Central Unit	34.5	SEER2:15.20
8. Skylights	Description	15. Heating Systems	kBtu/hr	Efficiency
U-Factor:(AVG)	N/A	a. Electric Heat Pump	34.0	HSPF2:7.80
SHGC(AVG):	N/A			
9. Floor Types	Insulation	16. Hot Water Systems		
a. Slab-On-Grade Edge Insulation	R= 0.0	a. Electric		Cap: 40 gallons
b. N/A	R=			EF: 0.950
c. N/A	R=	b. Conservation features		
				None
		17. Credits		CF, Pstat

I certify that this home has complied with the Florida Energy Efficiency Code for Building Construction through the above energy saving features which will be installed (or exceeded) in this home before final inspection. Otherwise, a new EPL Display Card will be completed based on installed Code compliant features.

Builder Signature: _____ Date: _____

Address of New Home: 3316 W Nassau Street

City/FL Zip: Tampa,FL,33607



*Note: This is not a Building Energy Rating. If your Index is below 70, your home may qualify for energy efficient mortgage (EEM) incentives if you obtain a Florida Energy Rating. For information about the Florida Building Code, Energy Conservation, contact the Florida Building Commission's support staff.

**Label required by Section R303.1.3 of the Florida Building Code, Energy Conservation, if not DEFAULT.

Residential System Sizing Calculation

Summary

Project Title:

3316 W Nassau Street
Tampa, FL 33607

3316 W Nassau Street 3 Ton 15 2

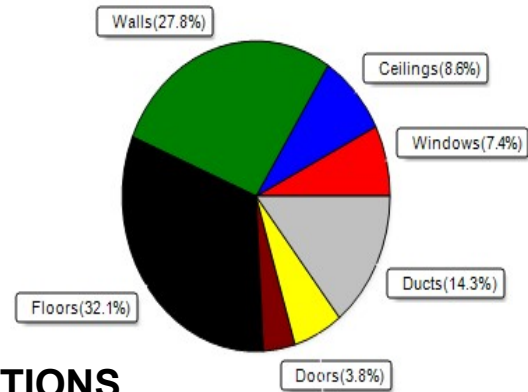
9/18/2025

Location for weather data: Tampa, FL - Defaults: Latitude(27.97) Altitude(10 ft.) Temp Range(M)					
Humidity data: Interior RH (50%) Outdoor wet bulb (77F) Humidity difference(54gr.)					
Winter design temperature(TMY3 99%) 36 F			Summer design temperature(TMY3 99%) 92 F		
Winter setpoint 70 F			Summer setpoint 75 F		
Winter temperature difference 34 F			Summer temperature difference 17 F		
Total heating load calculation 21215 Btuh			Total cooling load calculation 18392 Btuh		
Submitted heating capacity % of calc Btuh			Submitted cooling capacity % of calc Btuh		
Total (Electric Heat Pump) 160.3 34000			Sensible (SHR = 0.75) 186.9 25875		
Heat Pump + Auxiliary(0.0kW) 160.3 34000			Latent 189.7 8625		
			Total (Electric Heat Pump) 187.6 34500		

WINTER CALCULATIONS

Winter Heating Load (for 1634 sqft)

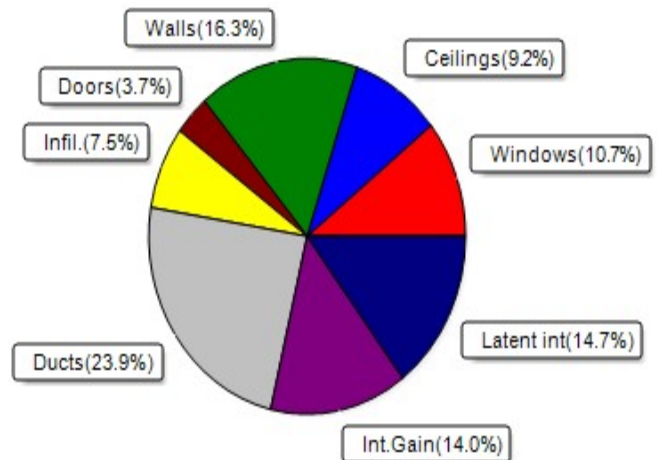
Load component	Load
Window total 136 sqft	1574 Btuh
Wall total 1164 sqft	5905 Btuh
Door total 60 sqft	816 Btuh
Ceiling total 1634 sqft	1818 Btuh
Floor total See detail report	6820 Btuh
Infiltration 33 cfm	1243 Btuh
Duct loss	3037 Btuh
Subtotal	21215 Btuh
Ventilation Ex:0 cfm; Sup:0 cfm	0 Btuh
TOTAL HEAT LOSS	21215 Btuh



SUMMER CALCULATIONS

Summer Cooling Load (for 1634 sqft)

Load component	Load
Window total 136 sqft	1969 Btuh
Wall total 1164 sqft	3005 Btuh
Door total 60 sqft	672 Btuh
Ceiling total 1634 sqft	1692 Btuh
Floor total	0 Btuh
Infiltration 25 cfm	466 Btuh
Internal gain	2580 Btuh
Duct gain	3460 Btuh
Sens.Ventilation Ex:0 cfm; Sup:0 cfm	0 Btuh
Blower Load	0 Btuh
Total sensible gain	13845 Btuh
Latent gain(ducts)	932 Btuh
Latent gain(infiltration)	916 Btuh
Latent gain(ventilation)	0 Btuh
Latent gain(internal/occupants/other)	2700 Btuh
Total latent gain	4548 Btuh
TOTAL HEAT GAIN	18392 Btuh



8th Edition

EnergyGauge® System Sizing
PREPARED BY: Navarrete Drafting Inc.
DATE: 9/18/25

Residential Window Diversity

MidSummer

Project Title:

3316 W Nassau Street
Tampa, FL 33607

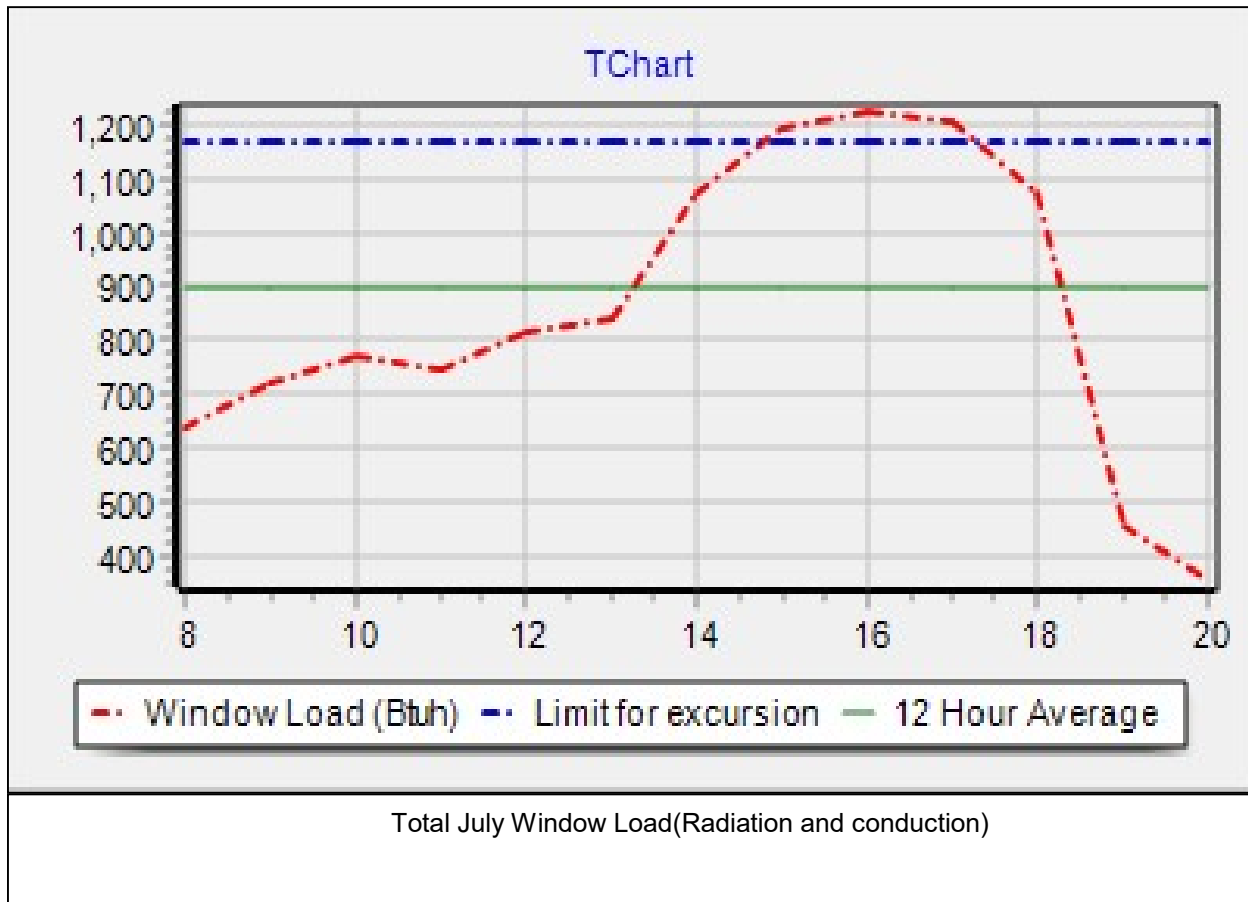
3316 W Nassau Street 3 Ton 15 2

9/18/2025

Weather data for: Tampa - Defaults

Summer design temperature	92 °F	Average window load for July	896 Btuh
Cooling setpoint	75 °F	Peak window load for July	1224 Btuh
Summer temperature difference	17 °F	Excursion limit(130% of Ave.)	1165 Btuh
Latitude	27.97 North	Window excursion (July)	58 Btuh

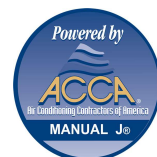
WINDOW Average and Peak Loads



Warning: This application has glass areas that produce relatively large heat gains for part of the day. Variable air volume devices may be required to overcome spikes in solar gain for one or more rooms. A zoned system may be required or some rooms may require zone control.

EnergyGauge® System Sizing for Florida residences only
PREPARED BY: Navarrete Drafting Inc.
DATE: 9/18/25

EnergyGauge® / USRCZB v8.1.00



Residential Window Diversity

October

Project Title:

3316 W Nassau Street
Tampa, FL 33607

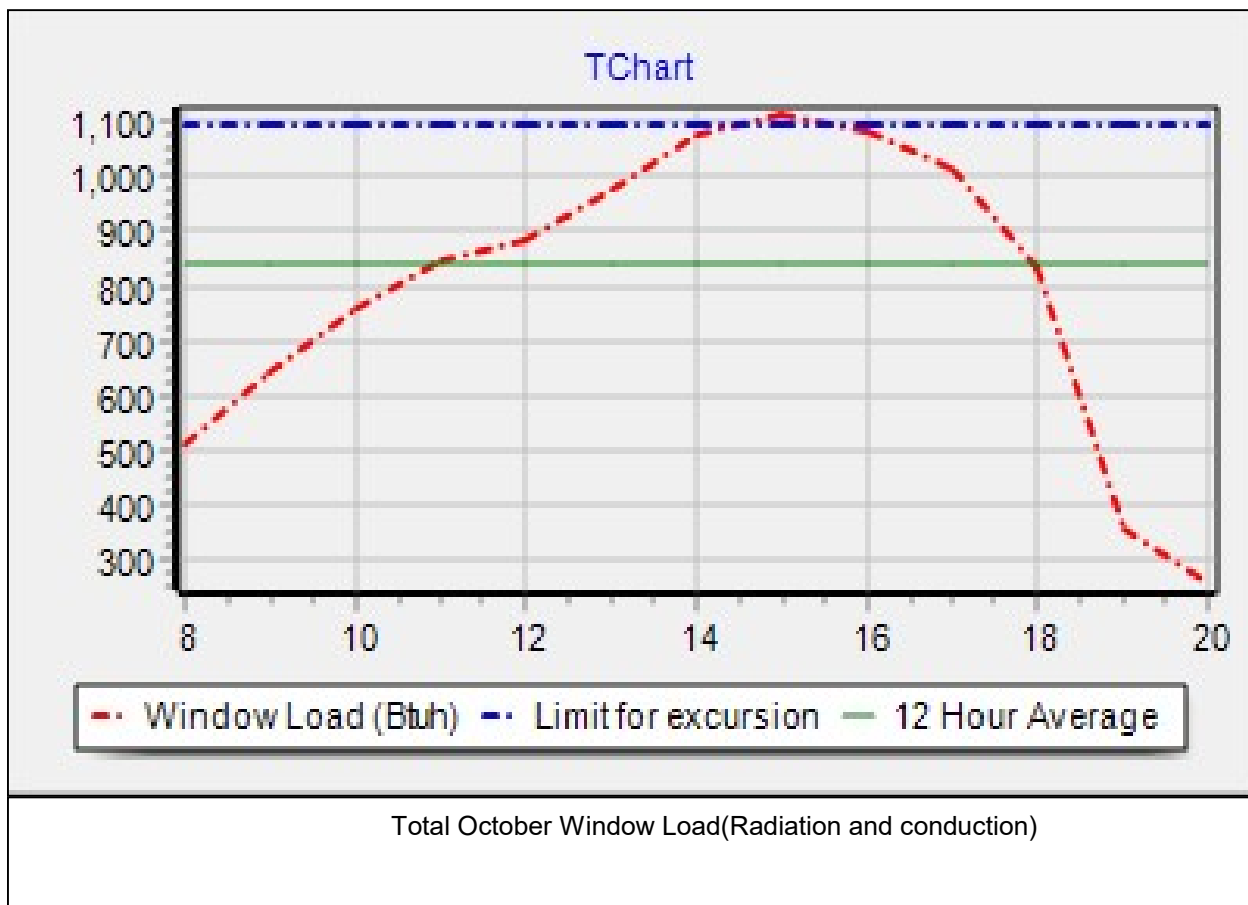
3316 W Nassau Street 3 Ton 15 2

9/18/2025

Weather data for: Tampa - Defaults

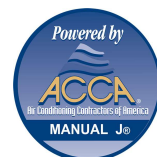
Summer design temperature	92 °F	Average window load for October	842 Btuh
Cooling setpoint	75 °F	Peak window load for October	1115 Btuh
Summer temperature difference	17 °F	Excursion limit(130% of Ave.)	1095 Btuh
Latitude	27.97 North	Window excursion (October)	20 Btuh

WINDOW Average and Peak Loads



Warning: This application has glass areas that produce relatively large heat gains for part of the day. Variable air volume devices may be required to overcome spikes in solar gain for one or more rooms. A zoned system may be required or some rooms may require zone control.

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DATE: 9/18/25



System Sizing Calculations - Winter

Residential Load - Whole House Component Details

3316 W Nassau Street
Tampa, FL 33607

Project Title:
3316 W Nassau Street 3 Ton 15 2
Building Type: User

9/18/2025

Reference City: Tampa, FL (Defaults) Winter Temperature Difference: 34.0 °F (TMY3 99%)
Winter Setpoint: 70 °F (Required Manual J default)

Component Loads for Whole House

Window	Panes/Type	Frame	U	Orientation	Area(sqft)	X	HTM=	Load
1	2, NFRC 0.23	Vinyl	0.34	N	9.3		11.6	107 Btuh
2	2, NFRC 0.23	Vinyl	0.34	N	18.5		11.6	214 Btuh
3	2, NFRC 0.23	Vinyl	0.34	E	37.0		11.6	428 Btuh
4	2, NFRC 0.23	Vinyl	0.34	S	18.5		11.6	214 Btuh
5	2, NFRC 0.23	Vinyl	0.34	W	27.8		11.6	321 Btuh
6	2, NFRC 0.23	Vinyl	0.34	W	6.7		11.6	77 Btuh
7	2, NFRC 0.23	Vinyl	0.34	N	18.5		11.6	214 Btuh
Window Total					136.2(sqft)			1574 Btuh
Walls	Type	Ornt.	Ueff.	R-Value (Cav/Sh)	Area	X	HTM=	Load
1	Conc Blk,Hollow	- Ext	(0.149)	4.1/0.0	115		5.07	582 Btuh
2	Conc Blk,Hollow	- Ext	(0.149)	4.1/0.0	46		5.07	231 Btuh
3	Conc Blk,Hollow	- Ext	(0.149)	4.1/0.0	315		5.07	1598 Btuh
4	Conc Blk,Hollow	- Ext	(0.149)	4.1/0.0	170		5.07	860 Btuh
5	Conc Blk,Hollow	- Ext	(0.149)	4.1/0.0	60		5.07	304 Btuh
6	Conc Blk,Hollow	- Ext	(0.149)	4.1/0.0	358		5.07	1814 Btuh
7	Conc Blk,Hollow	- Ext	(0.149)	4.1/0.0	62		5.07	312 Btuh
8	Conc Blk,Hollow	- Ext	(0.149)	4.1/0.0	40		5.07	203 Btuh
Wall Total					1164(sqft)			5905 Btuh
Doors	Type	Storm	Ueff.		Area	X	HTM=	Load
1	Insulated - Exterior,	n	(0.400)		20		13.6	272 Btuh
2	Insulated - Exterior,	n	(0.400)		20		13.6	272 Btuh
3	Insulated - Exterior,	n	(0.400)		20		13.6	272 Btuh
Door Total					60(sqft)			816Btuh
Ceilings	Type/Color/Surface		Ueff.	R-Value	Area	X	HTM=	Load
1	Flat ceil/D/Shing		(0.241)	0.0/23.0	1144		0.96	1103 Btuh
2	Single as/D/Shing		(0.043)	0.0/23.0	490		1.5	716 Btuh
Ceiling Total					1634(sqft)			1818Btuh
Floors	Type		Ueff.	R-Value	Size	X	HTM=	Load
1	Slab On Grade		(1.180)	0.0	96.0 ft(perim.)		40.1	3852 Btuh
2	Slab On Grade		(1.180)	0.0	74.0 ft(perim.)		40.1	2969 Btuh
Floor Total					1634 sqft			6820 Btuh
Envelope Subtotal:								16934 Btuh
Infiltration	Type	Wholehouse	ACH	Volume(cuft)	Wall Ratio	CFM=		Load
	Natural		0.15	13072	1.00	33.3		1243 Btuh
Duct load	Average sealed, R6.0, Supply(Att), Return(Att) (DLM of 0.167)							3037 Btuh

Manual J Winter Calculations

Residential Load - Component Details (continued)

3316 W Nassau Street
Tampa, FL 33607

Project Title:
3316 W Nassau Street 3 Ton 15 2
Building Type: User

9/18/2025

All Zones	Sensible Subtotal All Zones	21215 Btuh
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WHOLE HOUSE TOTALS

Totals for Heating	Subtotal Sensible Heat Loss Ventilation Sens. Heat Loss (Ex:0 cfm; Sup:0 cfm) Total Heat Loss	21215 Btuh 0 Btuh 21215 Btuh
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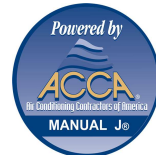
EQUIPMENT

1. Electric Heat Pump	#	34000 Btuh
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Key: Window types - NFRC (Requires U-Factor and Shading coefficient(SHGC) of glass as numerical values)
or - Glass as 'Clear' or 'Tint' (Uses U-Factor and SHGC defaults)

U - (Window U-Factor)

HTM - (ManualJ Heat Transfer Multiplier)



Version 8

System Sizing Calculations - Summer

Residential Load - Whole House Component Details

Project Title:

3316 W Nassau Street
Tampa, FL 33607

3316 W Nassau Street 3 Ton 15 2

9/18/2025

Reference City: Tampa, FL (Defaults)
Humidity difference: 54gr.

Temperature Difference: 17.0F(TMY3 99%)
Summer Setpoint: 75 °F (Required Manual J default)

Component Loads for Whole House

Window	Type*						Overhang		Window Area(sqft)			HTM		Load
	Panes	SHGC	U	InSh	IS	Ornt	Len	Hgt	Gross	Shaded	Unshaded	Shaded	Unshaded	
1	2 NFRC	0.23, 0.34	B-L	No	N		6.3ft	0.3ft	9.3	0.0	9.3	8	8	75 Btuh
2	2 NFRC	0.23, 0.34	B-L	No	N		1.0ft	2.3ft	18.5	0.0	18.5	8	8	149 Btuh
3	2 NFRC	0.23, 0.34	B-L	No	E		1.3ft	0.7ft	37.0	5.3	31.7	8	20	688 Btuh
4	2 NFRC	0.23, 0.34	B-L	No	S		1.0ft	3.3ft	18.5	18.5	0.0	8	9	149 Btuh
5	2 NFRC	0.23, 0.34	B-L	No	W		1.0ft	1.0ft	27.8	0.0	27.8	8	20	564 Btuh
6	2 NFRC	0.23, 0.34	B-L	No	W		1.0ft	1.0ft	6.7	0.0	6.7	8	20	136 Btuh
7	2 NFRC	0.23, 0.34	B-L	No	N		1.0ft	1.0ft	18.5	0.0	18.5	8	8	149 Btuh
	Excursion													58 Btuh
	Window Total								136 (sqft)					1969 Btuh
Walls	Type	U-Value				R-Value		Area(sqft)			HTM		Load	
							Cav/Sheath							
1	Concrete Blk,Hollow - Ext						0.15	4.1/0.0		114.8			2.6	296 Btuh
2	Concrete Blk,Hollow - Ext						0.15	4.1/0.0		45.5			2.6	117 Btuh
3	Concrete Blk,Hollow - Ext						0.15	4.1/0.0		315.0			2.6	813 Btuh
4	Concrete Blk,Hollow - Ext						0.15	4.1/0.0		169.5			2.6	438 Btuh
5	Concrete Blk,Hollow - Ext						0.15	4.1/0.0		60.0			2.6	155 Btuh
6	Concrete Blk,Hollow - Ext						0.15	4.1/0.0		357.6			2.6	923 Btuh
7	Concrete Blk,Hollow - Ext						0.15	4.1/0.0		61.5			2.6	159 Btuh
8	Concrete Blk,Hollow - Ext						0.15	4.1/0.0		40.0			2.6	103 Btuh
	Wall Total								1164 (sqft)					3005 Btuh
Doors	Type	U-Value				R-Value		Area (sqft)			HTM		Load	
1	Insulated - Exterior								20.0			11.2	224 Btuh	
2	Insulated - Exterior								20.0			11.2	224 Btuh	
3	Insulated - Exterior								20.0			11.2	224 Btuh	
	Door Total								60 (sqft)					672 Btuh
Ceilings	Type/Color/Surface	U-Value				R-Value		Area(sqft)			HTM		Load	
1	Unvented Attic/DarkShingle						0.241	0.0/23.0		1144.0			0.96	1103 Btuh
2	SnglAsmb w airsp/DarkShingle						0.043	0.0/23.0		490.0			1.20	589 Btuh
	Ceiling Total								1634 (sqft)					1692 Btuh
Floors	Type	U-Value				R-Value		Size			HTM		Load	
1	Slab On Grade								0.0			1144 (ft-perimeter)	0.0	0 Btuh
2	Slab On Grade								0.0			490 (ft-perimeter)	0.0	0 Btuh
	Floor Total								1634.0 (sqft)					0 Btuh
	Envelope Subtotal:													7338 Btuh

Manual J Summer Calculations

Residential Load - Component Details (continued)

3316 W Nassau Street
Tampa, FL 33607

Project Title: Climate:FL_TAMPA_INTERNATIONAL_AP
3316 W Nassau Street 3 Ton 15 2

9/18/2025

Infiltration	Type Natural	Average ACH 0.11	Volume(cuft) 13072	Wall Ratio 1	CFM= 24.9	Load 466 Btuh
Internal gain		Occupants 6	Btuh/occupant X 230	Appliance +	1200	Load 2580 Btuh
					Sensible Envelope Load:	10384 Btuh
Duct load	Average sealed, Supply(R6.0-Attic), Return(R6.0-Attic) (DGM of 0.333)					3460 Btuh
					Sensible Load All Zones	13845 Btuh

Manual J Summer Calculations

Residential Load - Component Details (continued)

3316 W Nassau Street
Tampa, FL 33607

Project Title: Climate:FL_TAMPA_INTERNATIONAL_AP
3316 W Nassau Street 3 Ton 15 2

9/18/2025

WHOLE HOUSE TOTALS

Whole House Totals for Cooling	Sensible Envelope Load All Zones	10384 Btuh
	Sensible Duct Load	3460 Btuh
	Total Sensible Zone Loads	13845 Btuh
	Sensible ventilation (Ex:0 cfm; Sup:0 cfm)	0 Btuh
	Blower	0 Btuh
	Total sensible gain	13845 Btuh
	Latent infiltration gain (for 54 gr. humidity difference)	916 Btuh
	Latent ventilation gain	0 Btuh
	Latent duct gain	932 Btuh
	Latent occupant gain (6.0 people @ 200 Btuh per person)	1200 Btuh
	Latent other gain	1500 Btuh
	Latent total gain	4548 Btuh
	TOTAL GAIN	18392 Btuh

EQUIPMENT

1. Central Unit	#	34500 Btuh
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*Key: Window types (Panels - Number and type of panes of glass)
(SHGC - Shading coefficient of glass as SHGC numerical value)
(U - Window U-Factor)
(InSh - Interior shading device: none(No), Blinds(B), Draperies(D) or Roller Shades(R))
- For Blinds: Assume medium color, half closed
- For Draperies: Assume medium weave, half closed
- For Roller shades: Assume translucent, half closed
(IS - Insect screen: none(N), Full(F) or Half(½))
(Ornt - compass orientation)



Version 8

RESIDENTIAL ENERGY CONSERVATION CODE DOCUMENTATION CHECKLIST

Florida Department of Business and Professional Regulation Simulated Performance Alternative (Performance) Method

Applications for compliance with the 2023 Florida Building Code, Energy Conservation via the Residential Simulated Performance Alternative shall include:

- ☐ *This checklist*
- ☐ *Form R405-2023 report*
- ☐ *Input summary checklist that can be used for field verification (usually four pages/may be greater)*
- ☐ *Energy Performance Level (EPL) Display Card (one page)*
- ☐ *HVAC system sizing and selection based on ACCA Manual S or per exceptions provided in Section R403.7*
- ☐ *Mandatory Requirements (five pages)*

Required prior to CO:

- ☐ *Air Barrier and Insulation Inspection Component Criteria checklist (Table R402.4.1.1 - one page)*
- ☐ *A completed 2023 Envelope Leakage Test Report (usually one page); exception in R402.4 allows dwelling units of R-2 Occupancies and multiple attached single family dwellings to comply with Section C402.5*
- ☐ *If Form R405 duct leakage type indicates anything other than "default leakage", then a completed 2023 Duct Leakage Test Report - Performance Method (usually one page)*

Florida Building Code, Energy Conservation, 8th Edition (2023)

Mandatory Requirements for Residential Performance, Prescriptive and ERI Methods

ADDRESS: 3316 W Nassau Street
Tampa, FL 33607

Permit Number:

MANDATORY REQUIREMENTS - See individual code sections for full details.

SECTION R401 GENERAL

- ☐ **R401.3 Energy Performance Level (EPL) display card - (Mandatory).** The building official shall require that an energy performance level (EPL) display card be completed and certified by the builder to be accurate and correct before final approval of the building for occupancy. Florida law (Section 553.9085, Florida Statutes) requires the EPL display card to be included as an addendum to each sales contract for both presold and nonpresold residential buildings. The EPL display card contains information indicating the energy performance level and efficiencies of components installed in a dwelling unit. The building official shall verify that the EPL display card completed and signed by the builder accurately reflects the plans and specifications submitted to demonstrate code compliance for the building. A copy of the EPL display card can be found in Appendix RD.

SECTION R402 BUILDING THERMAL ENVELOPE

- ☐ **R402.2.10.1 Slab-on-grade floor insulation installation (Mandatory).** Where installed, the insulation shall extend downward from the top of the slab on the outside or inside of the foundation wall. Insulation located below grade shall be extended the distance provided in Table R402.1.2, or the distance of the proposed design as applicable, by any combination of vertical insulation, insulation extending under the slab or insulation extending out from the building. Insulation extending away from the building shall be protected by pavement or by not less than 10 inches (254 mm) of soil. The top edge of the insulation installed between the exterior wall and the edge of the interior slab shall be permitted to be cut at a 45-degree (0.79 rad) angle away from the exterior wall.
- ☐ **R402.2.11.1 Crawl space walls insulation installation (Mandatory).** Where crawl space wall insulation is installed, it shall be permanently fastened to the wall and extend downward from the floor to the finished grade level and then vertically and/or horizontally for at least an additional 24 inches (610 mm). Exposed earth in unvented crawl space foundations shall be covered with a continuous Class I vapor retarder in accordance with the Florida Building Code, Building, or Florida Building Code, Residential, as applicable. All joints of the vapor retarder shall overlap by 6 inches (153 mm) and be sealed or taped. The edges of the vapor retarder shall extend not less than 6 inches (153 mm) up the stem wall and shall be attached to the stem wall.
- ☐ **R402.4 Air leakage (Mandatory).** The building thermal envelope shall be constructed to limit air leakage in accordance with the requirements of Sections R402.4.1 through R402.4.5.
 - Exception:** Dwelling units of R-2 Occupancies and multiple attached single family dwellings shall be permitted to comply with Section C402.5.
- ☐ **R402.4.1 Building thermal envelope.** The building thermal envelope shall comply with Sections R402.4.1.1 and R402.4.1.2. The sealing methods between dissimilar materials shall allow for differential expansion and contraction.
- ☐ **R402.4.1.1 Installation.** The components of the building thermal envelope as listed in Table R402.4.1.1 shall be installed in accordance with the manufacturer's instructions and the criteria listed in Table R402.4.1.1, as applicable to the method of construction. Where required by the code official, an approved third party shall inspect all components and verify compliance.
- ☐ **R402.4.1.2 Testing.** The building or dwelling unit shall be tested and verified as having an air leakage rate not exceeding seven air changes per hour in Climate Zones 1 and 2, and three air changes per hour in Climate Zones 3 through 8. Dwelling units with an air leakage rate less than three air changes per hour shall be provided with whole-house mechanical ventilation in accordance with Section R403.6.1 of this code and Section M1507.3 of the Florida Building Code, Residential. Testing shall be conducted in accordance with ANSI/RESNET/ICC 380 and reported at a pressure of 0.2 inch w.g. (50 pascals). Testing shall be conducted by either individuals as defined in Section 553.993(5) or (7), Florida Statutes, or individuals licensed as set forth in Section 489.105(3)(f), (g) or (i) or an approved third party. A written report of the results of the test shall be signed by the party conducting the test and provided to the code official. Testing shall be performed at any time after creation of all penetrations of the building thermal envelope.
 - Exception:** Testing is not required for additions, alterations, renovations, or repairs, of the building thermal envelope of existing buildings in which the new construction is less than 85 percent of the building thermal envelope.

During testing:

1. Exterior windows and doors, fireplace and stove doors shall be closed, but not sealed, beyond the intended weatherstripping or other infiltration control measures.
2. Dampers including exhaust, intake, makeup air, backdraft and flue dampers shall be closed, but not sealed beyond intended infiltration control measures.
3. Interior doors, if installed at the time of the test, shall be open.
4. Exterior doors for continuous ventilation systems and heat recovery ventilators shall be closed and sealed.
5. Heating and cooling systems, if installed at the time of the test, shall be turned off.
6. Supply and return registers, if installed at the time of the test, shall be fully open.
7. If an attic is both air sealed and insulated at the roof deck, interior access doors and hatches between the conditioned space volume and the attic shall be opened during the test and the volume of the attic shall be added to the conditioned space volume for purposes of reporting an infiltration volume and calculating the air leakage of the home.

Florida Building Code, Energy Conservation, Mandatory Requirements (2023 Continued)

- ☐ **R402.4.2 Fireplaces.** New wood-burning fireplaces shall have tight-fitting flue dampers or doors, and outdoor combustion air. Where using tight-fitting doors on factory-built fireplaces listed and labeled in accordance with UL 127, the doors shall be tested and listed for the fireplace. Where using tight-fitting doors on masonry fireplaces, the doors shall be listed and labeled in accordance with UL 907.
- ☐ **R402.4.3 Fenestration air leakage.** Windows, skylights and sliding glass doors shall have an air infiltration rate of no more than 0.3 cfm per square foot (1.5 L/s/m²), and swinging doors no more than 0.5 cfm per square foot (2.6 L/s/m²), when tested according to NFRC 400 or AAMA/WDMA/CSA 101/I.S.2/A440 by an accredited, independent laboratory and listed and labeled by the manufacturer.

Exception: Site-built windows, skylights and doors.
- ☐ **R402.4.4 Rooms containing fuel - burning appliances.** In Climate Zones 3 through 8, where open combustion air ducts provide combustion air to open combustion fuel burning appliances, the appliances and combustion air opening shall be located outside the building thermal envelope or enclosed in a room, isolated from inside the thermal envelope. Such rooms shall be sealed and insulated in accordance with the envelope requirements of Table R402.1.2, where the walls, floors and ceilings shall meet not less than the basement wall R-value requirement. The door into the room shall be fully gasketed and any water lines and ducts in the room insulated in accordance with Section R403. The combustion air duct shall be insulated where it passes through conditioned space to a minimum of R-8.

Exceptions:

 1. Direct vent appliances with both intake and exhaust pipes installed continuous to the outside.
 2. Fireplaces and stoves complying with Section R402.4.2 and Section R1006 of the Florida Building Code, Residential.
- ☐ **R402.4.5 Recessed lighting.** Recessed luminaires installed in the building thermal envelope shall be sealed to limit air leakage between conditioned and unconditioned spaces. All recessed luminaires shall be IC-rated and labeled as having an air leakage rate not more than 2.0 cfm (0.944 L/s) when tested in accordance with ASTM E283 at a 1.57 psf (75 Pa) pressure differential. All recessed luminaires shall be sealed with a gasket or caulk between the housing and the interior wall or ceiling covering.
- ☐ **R402.4.6 Air-sealed electrical and communication boxes.** Air-sealed electrical and communication boxes that penetrate the air barrier of the building thermal envelope shall be caulked, taped, gasketed, or otherwise sealed to the air barrier element being penetrated. Air-sealed boxes shall be buried in or surrounded by insulation. Air-sealed boxes shall be marked in accordance with NEMA OS 4. Air-sealed boxes shall be installed in accordance with the manufacturer's instructions.

SECTION R403 SYSTEMS

R403.1 Controls

- ☐ **R403.1.1 Thermostat provision (Mandatory).** At least one thermostat shall be provided for each separate heating and cooling system
- ☐ **R403.1.3 Heat pump supplementary heat (Mandatory).** Heat pumps with supplementary electric-resistance heaters shall have controls that limit supplemental heat operation to only those times when one of the following applies:
 1. The vapor compression cycle cannot provide the necessary heating energy to satisfy the thermostat setting.
 2. The heat pump is operating in defrost mode.
 3. The vapor compression cycle malfunctions.
 4. The thermostat malfunctions
- ☐ **R403.3.2 Sealing (Mandatory).** All ducts, air handlers, filter boxes and building cavities that form the primary air containment passageways for air distribution systems shall be considered ducts or plenum chambers, shall be constructed and sealed in accordance with Section C403.2.9.2 of the Commercial Provisions of this code and shall be shown to meet duct tightness criteria below.

Duct tightness shall be verified by testing in accordance with ANSI/RESNET/ICC 380 by either individuals as defined in Section 553.993(5) or (7), Florida Statutes, or individuals licensed as set forth in Section 489.105(3)(f), (g) or (i), Florida Statutes, to be "substantially leak free" in accordance with Section R403.3.3.
- ☐ **R403.3.2.1 Sealed air handler.** Air handlers shall have a manufacturer's designation for an air leakage of no more than 2 percent of the design airflow rate when tested in accordance with ASHRAE 193.
- ☐ **R403.3.3 Duct testing (Mandatory).** Ducts shall be pressure tested to determine air leakage by one of the following methods:
 1. Rough-in test: Total leakage shall be measured with a pressure differential of 0.1 inch w.g. (25 Pa) across the system, including the manufacturer's air handler enclosure if installed at the time of the test. All registers shall be taped or otherwise sealed during the test.
 2. Postconstruction test: Total leakage shall be measured with a pressure differential of 0.1 inch w.g. (25 Pa) across the entire system, including the manufacturer's air handler enclosure. Registers shall be taped or otherwise sealed during the test.

Exceptions;

 1. A duct air leakage test shall not be required where the ducts and air handlers are located entirely within the building thermal envelope.
 2. Duct testing is not mandatory for buildings complying by Section 405 of this code. Duct leakage testing is required for Section R405 compliance where credit is taken for leakage, and a duct air leakage Q_n to the outside of less than 0.080 (where Q_n = duct leakage to the outside in cfm per 100 square feet of conditioned floor area tested at 25 Pascals) is indicated in the compliance report for the proposed design.

A written report of the results of the test shall be signed by the party conducting the test and provided to the code official

Florida Building Code, Energy Conservation, Mandatory Requirements (2023 Continued)

- ☐ **R403.3.5 Building cavities (Mandatory).** Building framing cavities shall not be used as ducts or plenums
- ☐ **R403.4 Mechanical system piping insulation (Mandatory).** Mechanical system piping capable of carrying fluids above 105°F (41°C) or below 55°F (13°C) shall be insulated to a minimum of R-3.
- ☐ **R403.4.1 Protection of piping insulation.** Piping insulation exposed to weather shall be protected from damage, including that caused by sunlight, moisture, equipment maintenance and wind, and shall provide shielding from solar radiation that can cause degradation of the material. Adhesive tape shall not be permitted.
- ☐ **R403.5.1 Heated water circulation and temperature maintenance systems (Mandatory).** If heated water circulation systems are installed, they shall be in accordance with Section R403.5.1.1. Heat trace temperature maintenance systems shall be in accordance with Section R403.5.1.2. Automatic controls, temperature sensors and pumps shall be accessible. Manual controls shall be readily accessible.
- ☐ **R403.5.1.1 Circulation systems.** Heated water circulation systems shall be provided with a circulation pump. The system return pipe shall be a dedicated return pipe or a cold water supply pipe. Gravity and thermosiphon circulation systems shall be prohibited. Controls for circulating hot water system pumps shall start the pump based on the identification of a demand for hot water within the occupancy. The controls shall automatically turn off the pump when the water in the circulation loop is at the desired temperature and when there is no demand for hot water.
- ☐ **R403.5.1.2 Heat trace systems.** Electric heat trace systems shall comply with IEEE 515.1 or UL 515. Controls for such systems shall automatically adjust the energy input to the heat tracing to maintain the desired water temperature in the piping in accordance with the times when heated water is used in the occupancy.
- ☐ **R403.5.2 Demand recirculation water systems (Mandatory).** Where installed, demand recirculation water systems shall have controls that comply with both of the following:
 - 1. The control shall start the pump upon receiving a signal from the action of a user of a fixture or appliance, sensing the presence of a user of a fixture or sensing the flow of hot or tempered water to a fixture fitting or appliance.
 - 2. The control shall limit the temperature of the water entering the cold water piping to 104°F (40°C).
- ☐ **R403.5.5 Heat traps (Mandatory).** Storage water heaters not equipped with integral heat traps and having vertical pipe risers shall have heat traps installed on both the inlets and outlets. External heat traps shall consist of either a commercially available heat trap or a downward and upward bend of at least 3 ½ inches (89 mm) in the hot water distribution line and cold water line located as close as possible to the storage tank.
- ☐ **R403.5.6 Water heater efficiencies (Mandatory).**
 - ☐ **R403.5.6.1.1 Automatic controls.** Service water-heating systems shall be equipped with automatic temperature controls capable of adjustment from the lowest to the highest acceptable temperature settings for the intended use. The minimum temperature setting range shall be from 100°F to 140°F (38°C to 60°C).
 - ☐ **R403.5.6.1.2 Shut down.** A separate switch or a clearly marked circuit breaker shall be provided to permit the power supplied to electric service systems to be turned off. A separate valve shall be provided to permit the energy supplied to the main burner(s) of combustion types of service water-heating systems to be turned off.
 - ☐ **R403.5.6.2 Water-heating equipment.** Water-heating equipment installed in residential units shall meet the minimum efficiencies of Table C404.2 in Chapter 4 of the Florida Building Code, Energy Conservation, Commercial Provisions, for the type of equipment installed. Equipment used to provide heating functions as part of a combination system shall satisfy all stated requirements for the appropriate water-heating category. Solar water heaters shall meet the criteria of Section R403.5.6.2.1.
 - ☐ **R403.5.6.2.1 Solar water-heating systems.** Solar systems for domestic hot water production are rated by the annual solar energy factor of the system. The solar energy factor of a system shall be determined from the Florida Solar Energy Center Directory of Certified Solar Systems. Solar collectors shall be tested in accordance with ISO Standard 9806, Test Methods for Solar Collectors, and SRCC Standard TM-1, Solar Domestic Hot Water System and Component Test Protocol. Collectors in installed solar water-heating systems should meet the following criteria:
 - 1. Be installed with a tilt angle between 10 degrees and 40 degrees of the horizontal; and
 - 2. Be installed at an orientation within 45 degrees of true south.
- ☐ **R403.6 Mechanical ventilation (Mandatory).** The building shall be provided with ventilation that meets the requirements of the Florida Building Code, Residential, or Florida Building Code, Mechanical, as applicable, or with other approved means of ventilation including: Natural, Infiltration or Mechanical means. Outdoor air intakes and exhausts shall have automatic or gravity dampers that close when the ventilation system is not operating.

Florida Building Code, Energy Conservation, Mandatory Requirements (2023 Continued)

- ☐ **R403.6.1 Whole-house mechanical ventilation system fan efficacy.** When installed to function as a whole-house mechanical ventilation system, fans shall meet the efficacy requirements of Table R403.6.1.
- Exception:** Where an air handler that is integral to tested and listed HVAC equipment is used to provide whole-house mechanical ventilation, the air handler shall be powered by an electronically commutated motor.

TABLE R403.6.1
WHOLE-HOUSE MECHANICAL VENTILATION SYSTEM FAN EFFICACY

FAN LOCATION	AIRFLOW RATE MINIMUM (CFM)	MINIMUM EFFICACY ^a (CFM/WATT)	AIRFLOW RATE MAXIMUM (CFM)
HRV or ERV	Any	1.2 cfm/watt	Any
Range hoods	Any	2.8 cfm/watt	Any
In-line fan	Any	3.8 cfm/watt	Any
Bathroom, utility room	10	2.8 cfm/watt	<90
Bathroom, utility room	90	3.5 cfm/watt	Any

For SI: 1 cfm = 28.3 L/min.

a. When tested in accordance with HVI Standard 916

- ☐ **R403.6.2 Ventilation Air.** Residential buildings designed to be operated at a positive indoor pressure or for mechanical ventilation shall meet the following criteria:
1. The design air change per hour minimums for residential buildings in ASHRAE 62.2, Ventilation for Acceptable Indoor Air Quality, shall be the maximum rates allowed for residential applications.
 2. No ventilation or air-conditioning system make-up air shall be provided to conditioned space from attics, crawlspaces, attached enclosed garages or outdoor spaces adjacent to swimming pools or spas.
 3. If ventilation air is drawn from enclosed space(s), then the walls of the space(s) from which air is drawn shall be insulated to a minimum of R-11 and the ceiling shall be insulated to a minimum of R-19, space permitting, or R-10 otherwise.

R403.7 Heating and cooling equipment.

- ☐ **R403.7.1 Equipment sizing (Mandatory).** Heating and cooling equipment shall be sized in accordance with ACCA Manual S based on the equipment loads calculated in accordance with ACCA Manual J or other approved heating and cooling calculation methodologies, based on building loads for the directional orientation of the building. The manufacturer and model number of the outdoor and indoor units (if split system) shall be submitted along with the sensible and total cooling capacities at the design conditions described in Section R302.1. This Code does not allow designer safety factors, provisions for future expansion or other factors that affect equipment sizing. System sizing calculations shall not include loads created by local intermittent mechanical ventilation such as standard kitchen and bathroom exhaust systems. New or replacement heating and cooling equipment shall have an efficiency rating equal to or greater than the minimum required by federal law for the geographic location where the equipment is installed.

Florida Building Code, Energy Conservation, Mandatory Requirements (2023 Continued)

- ☐ **R403.7.1.1 Cooling equipment capacity.** Cooling only equipment shall be selected so that its total capacity is not less than the calculated total load but not more than 1.15 times greater than the total load calculated according to the procedure selected in Section R403.7, or the closest available size provided by the manufacturer's product lines. The corresponding latent capacity of the equipment shall not be less than the calculated latent load.
- The published value for AHRI total capacity is a nominal, rating-test value and shall not be used for equipment sizing. Manufacturer's expanded performance data shall be used to select cooling-only equipment. This selection shall be based on the outdoor design dry-bulb temperature for the load calculation (or entering water temperature for water-source equipment), the blower CFM provided by the expanded performance data, the design value for entering wet-bulb temperature and the design value for entering dry-bulb temperature.

Design values for entering wet-bulb and dry-bulb temperatures shall be for the indoor dry bulb and relative humidity used for the load calculation and shall be adjusted for return side gains if the return duct(s) is installed in an unconditioned space.

Exceptions:

1. Attached single- and multiple-family residential equipment sizing may be selected so that its cooling capacity is less than the calculated total sensible load but not less than 80 percent of that load.
2. When signed and sealed by a Florida-registered engineer, in attached single- and multiple-family units, the capacity of equipment may be sized in accordance with good design practice.

R403.7.1.2 Heating equipment capacity.

- ☐ **R403.7.1.2.1 Heat pumps.** Heat pump sizing shall be based on the cooling requirements as calculated according to Section R403.7.1.1, and the heat pump total cooling capacity shall not be more than 1.15 times greater than the design cooling load even if the design heating load is 1.15 times greater than the design cooling load.
- ☐ **R403.7.1.2.2 Electric resistance furnaces.** Electric resistance furnaces shall be sized within 4 kW of the design requirements calculated according to the procedure selected in Section R403.7.1.
- ☐ **R403.7.1.2.3 Fossil fuel heating equipment.** The capacity of fossil fuel heating equipment with natural draft atmospheric burners shall not be less than the design load calculated in accordance with Section R403.7.1.
- ☐ **R403.7.1.3 Extra capacity required for special occasions.** Residences requiring excess cooling or heating equipment capacity on an intermittent basis, such as anticipated additional loads caused by major entertainment events, shall have equipment sized or controlled to prevent continuous space cooling or heating within that space by one or more of the following options:
1. A separate cooling or heating system is utilized to provide cooling or heating to the major entertainment areas.
 2. A variable capacity system sized for optimum performance during base load periods is utilized.
- ☐ **R403.8 Systems serving multiple dwelling units (Mandatory).** Systems serving multiple dwelling units shall comply with Sections C403 and C404 of the Florida Building Code, Energy Conservation—Commercial Provisions in lieu of Section R403.
- ☐ **R403.9 Snow melt and ice system controls (Mandatory).** Snow- and ice-melting systems, supplied through energy service to the building, shall include automatic controls capable of shutting off the system when the pavement temperature is above 50°F (10°C), and no precipitation is falling and an automatic or manual control that will allow shutoff when the outdoor temperature is above 40°F (4.8°C).
- ☐ **403.10 Pools and permanent spa energy consumption (Mandatory).** The energy consumption of pools and permanent spas shall be in accordance with Sections R403.10.1 through R403.10.5.
- ☐ **R403.10.1 Heaters.** The electric power to heaters shall be controlled by a readily accessible on-off switch that is an integral part of the heater mounted on the exterior of the heater, or external to and within 3 feet (914 mm) of the heater. Operation of such switch shall not change the setting of the heater thermostat. Such switches shall be in addition to a circuit breaker for the power to the heater.
- Gas-fired heaters shall not be equipped with continuously burning ignition pilots.
- ☐ **R403.10.2 Time switches.** Time switches or other control methods that can automatically turn off and on according to a preset schedule shall be installed for heaters and pump motors. Heaters and pump motors that have built-in time switches shall be in compliance with this section.
- Exceptions:**
1. Where public health standards require 24-hour pump operation.
 2. Pumps that operate solar- and waste-heat-recovery pool heating systems
 3. Where pumps are powered exclusively from on-site renewable generation.

Florida Building Code, Energy Conservation, Mandatory Requirements (2023 Continued)

- ☐ **R403.10.3 Covers.** Outdoor heated swimming pools and outdoor permanent spas shall be equipped with a vapor-retardant cover on or at the water surface or a liquid cover or other means proven to reduce heat loss.
- Exception:** Where more than 70 percent of the energy for heating, computed over an operation season, is from site-recovered energy, such as from a heat pump or solar energy source, covers or other vapor-retardant means shall not be required.
- ☐ **R403.10.4 Gas- and oil-fired pool and spa heaters.** All gas- and oil-fired pool and spa heaters shall have a minimum thermal efficiency of 82 percent for heaters manufactured on or after April 16, 2013, when tested in accordance with ANSI Z 21.56. Pool heaters fired by natural or LP gas shall not have continuously burning pilot lights.
- ☐ **R403.10.5 Heat pump pool heaters.** Heat pump pool heaters shall have a minimum COP of 4.0 when tested in accordance with AHRI 1160, Table 2, Standard Rating Conditions-Low Air Temperature. A test report from an independent laboratory is required to verify procedure compliance. Geothermal swimming pool heat pumps are not required to meet this standard.
- ☐ **R403.11 Portable spas (Mandatory).** The energy consumption of electric-powered portable spas shall be controlled by the requirements of APSP-14
- ☐ **R403.13 Dehumidifiers (Mandatory).** If installed, a dehumidifier shall conform to the following requirements:
1. The minimum rated efficiency of the dehumidifier shall be greater than 1.7 liters/ kWh if the total dehumidifier capacity for the house is less than 75 pints/day and greater than 2.38 liters/kWh if the total dehumidifier capacity for the house is greater than or equal to 75 pints/day.
 2. The dehumidifier shall be controlled by a sensor that is installed in a location where it is exposed to mixed house air.
 3. Any dehumidifier unit located in unconditioned space that treats air from conditioned space shall be insulated to a minimum of R-2.
 4. Condensate disposal shall be in accordance with Section M1411.3.1 of the Florida Building Code, Residential.
- ☐ **R403.13.1 Ducted dehumidifiers.** Ducted dehumidifiers shall, in addition to conforming to the requirements of Section R403.13, conform to the following requirements:
1. If a ducted dehumidifier is configured with return and supply ducts both connected into the supply side of the cooling system, a backdraft damper shall be installed in the supply air duct between the dehumidifier inlet and outlet duct.
 2. If a ducted dehumidifier is configured with only its supply duct connected into the supply side of the central heating and cooling system, a backdraft damper shall be installed in the dehumidifier supply duct between the dehumidifier and central supply duct.
 3. A ducted dehumidifier shall not be ducted to or from a central ducted cooling system on the return duct side upstream from the central cooling evaporator coil.
 4. Ductwork associated with a dehumidifier located in unconditioned space shall be insulated to a minimum of R-6.

SECTION R404 ELECTRICAL POWER AND LIGHTING SYSTEMS

- ☐ **R404.1 Lighting equipment (Mandatory).** All permanently installed luminaires, excluding those in kitchen appliances, shall have an efficacy of at least 45 lumens-per-watt or shall utilize lamps with an efficacy of not less than 65 lumens-per-watt.
- R404.1.1 Lighting equipment (Mandatory).** Fuel gas lighting systems shall not have continuously burning pilot lights.

SECTION R405 SIMULATED PERFORMANCE ALTERNATIVE (PERFORMANCE)

- ☐ **R405.2 Mandatory requirements.** Compliance with this section requires that the mandatory provisions identified in Section R401.2 be met. All supply and return ducts not completely inside the building thermal envelope shall be insulated to a minimum of R-6, except site-wrapped supply ducts not completely inside the building thermal envelope shall be insulated to a minimum of R-8.
- ☐ **R405.2.1 Ceiling insulation.** Ceilings shall have an insulation level of at least R-19, space permitting. For the purposes of this code, types of ceiling construction that are considered to have inadequate space to install R-19 include single assembly ceilings of the exposed deck and beam type and concrete deck roofs. Such ceiling assemblies shall be insulated to at least a level of R-10.
- ☐ **R405.2.2 Building air leakage testing.** Building or dwelling air leakage testing shall be in accordance with Sections R402.4 through R402.4.1.2. If an air leakage rate below seven air changes per hour at a pressure of 0.2 inch w.g. (50 pascals) is specified for the proposed design, testing shall verify the air leakage rate does not exceed the air leakage rate of the proposed design instead of seven air changes per hour.
- ☐ **R405.2.3 Duct air leakage testing.** In cases where duct air leakage lower than the default Q_n to outside of 0.080 (where Q_n = duct leakage to the outside in cfm per 100 square feet of conditioned floor area tested at 25 Pascals) is specified for the proposed design, testing in accordance with Section R403.3.2 shall verify a duct air leakage rate not exceeding the leakage rate of the proposed design. Otherwise, in accordance with Section R403.3.3, duct testing is not mandatory for buildings complying by Section R405.

SECTION R406 ENERGY RATING INDEX COMPLIANCE ALTERNATIVE

- ☐ **R406.2 Mandatory requirements.** Compliance with this section requires that the provisions identified in Sections R401 through R404 labeled as “mandatory” and Section R403.5.3 of the 2015 International Energy Conservation Code be met. For buildings that do not utilize on-site renewable power production for compliance with this section, the building thermal envelope shall be greater than or equal to levels of efficiency and Solar Heat Gain Coefficient in Table 402.1.1 or 402.1.3 of the 2009 International Energy Conservation Code. For buildings that utilize on-site renewable power production for compliance with this section, the building thermal envelope shall be greater than or equal to levels of efficiency and Solar Heat Gain Coefficient in Table R402.1.2 or Table R402.1.4 of the 2015 International Energy Conservation Code.

Exception: Supply and return ducts not completely inside the building thermal envelope shall be insulated to a minimum of R-6.
- ☐ **R406.2.1 Site-wrapped supply ducts.** Site-wrapped supply ducts not completely inside the building thermal envelope shall be insulated to a minimum of R-8.

2023 - AIR BARRIER AND INSULATION INSPECTION COMPONENT CRITERIA-TABLE 402.4.1.1 a

Project Name: 3316 W Nassau Street 3 Ton 15 2		Builder Name:	
Street: 3316 W Nassau Street		Permit Office:	
City, State, Zip: Tampa, FL, 33607		Permit Number:	
Owner:		Jurisdiction:	
Design Location: FL, Tampa		County: Hillsborough(Florida Climate Zone 2)	
COMPONENT	AIR BARRIER CRITERIA	INSULATION INSTALLATION CRITERIA	ICC
General requirements	A continuous air barrier shall be installed in the building envelope. The exterior thermal envelope contains a continuous air barrier. Breaks or joints in the air barrier shall be sealed.	Air-permeable insulation shall not be used as a sealing material.	IC
Ceiling/attic	The air barrier in any dropped ceiling/soffit shall be aligned with the insulation and any gaps in the air barrier shall be sealed. Access openings, drop down stairs or knee wall doors to unconditioned attic spaces shall be sealed.	The insulation in any dropped ceiling/soffit shall be aligned with the air barrier.	
Walls	The junction of the foundation and sill plate shall be sealed. The junction of the top plate and the top of exterior walls shall be sealed. Knee walls shall be sealed.	Cavities within corners and headers of frame walls shall be insulated by completely filling the cavity with a material having a thermal resistance of R-3 per inch minimum. Exterior thermal envelope insulation for framed walls shall be installed in substantial contact and continuous alignment with the air barrier.	
Windows, skylights and doors	The space between window/door jambs and framing, and skylights and framing shall be sealed.		
Rim joists	Rim joists shall include the air barrier.	Rim joists shall be insulated.	
Floors (including above-garage and cantilevered floors)	The air barrier shall be installed at any exposed edge of insulation.	Floor framing cavity insulation shall be installed to maintain permanent contact with the underside of subfloor decking, or floor framing cavity insulation shall be permitted to be in contact with the top side of sheathing, or continuous insulation installed on the underside of floor framing and extends from the bottom to the top of all perimeter floor framing members.	
Crawl space walls	Exposed earth in unvented crawl spaces shall be covered with a Class I vapor retarder with overlapping joints taped.	Where provided instead of floor insulation, insulation shall be permanently attached to the crawlspace walls.	
Shafts, penetrations	Duct shafts, utility penetrations, and flue shafts opening to exterior or unconditioned space shall be sealed.		
Narrow cavities		Batts in narrow cavities shall be cut to fit, or narrow cavities shall be filled by insulation that on installation readily conforms to the available cavity spaces.	
Garage separation	Air sealing shall be provided between the garage and conditioned spaces.		
Recessed lighting	Recessed light fixtures installed in the building thermal envelope shall be sealed to the finished surface.	Recessed light fixtures installed in the building thermal envelope shall be air tight and IC rated.	
Plumbing and wiring		Batt insulation shall be cut neatly to fit around wiring and plumbing in exterior walls, or insulation that on installation readily conforms to available space shall extend behind piping and wiring.	
Shower/tub on exterior wall	The air barrier installed at exterior walls adjacent to showers and tubs shall separate them from the showers and tubs.	Exterior walls adjacent to showers and tubs shall be insulated.	
Electrical, communication, and other equipment boxes, housings, and enclosures	Boxes, housings, and enclosures that penetrate the air barrier shall be caulked, taped, gasketed, or otherwise sealed to the air barrier element being penetrated. All concealed openings into the box, housing, or enclosure shall be sealed. The continuity of the air barrier shall be maintained around boxes, housings, and enclosures that penetrate the air barrier. Alternatively, air-sealed boxes shall be installed in accordance with R402.4.6	Boxes, housings, and enclosures shall be buried in or surrounded by tightly fitted insulation.	
HVAC register boots	HVAC supply and return register boots that penetrate building thermal envelope shall be sealed to the sub-floor, wall covering or ceiling penetrated by the boot.		
Concealed sprinklers	When required to be sealed, concealed fire sprinklers shall only be sealed in a manner that is recommended by the manufacturer. Caulking or other adhesive sealants shall not be used to fill voids between fire sprinkler cover plates and walls or ceilings.		

a. In addition, inspection of log walls shall be in accordance with the provisions of ICC-400.

Envelope Leakage Test Report (Blower Door Test)
Residential Prescriptive, Performance or ERI Method Compliance
2023 Florida Building Code, Energy Conservation, 8th Edition

Jurisdiction:	Permit #:		
Job Information			
Builder:	Community:	Lot:	NA
Address: 3316 W Nassau Street			
City: Tampa	State: FL	Zip:	33607
Air Leakage Test Results <i>Passing results must meet either the Performance, Prescriptive, or ERI Method</i>			
☐ PRESCRIPTIVE METHOD-The building or dwelling unit shall be tested and verified as having an air leakage rate of not exceeding 7 air changes per hour at a pressure of 0.2 inch w.g. (50 Pascals) in Climate Zones 1 and 2. ☒ PERFORMANCE or ERI METHOD-The building or dwelling unit shall be tested and verified as having an air leakage rate of not exceeding the selected ACH(50) value, as shown on Form R405-2023 (Performance) or R406-2023 (ERI), section labeled as infiltration, sub-section ACH50. ACH(50) specified on Form R405-2023-Energy Calc (Performance) or R406-2023 (ERI): 5.000			
$\frac{\text{CFM}(50)}{\text{Building Volume}} \times 60 \div \frac{13072}{\text{ACH}(50)} =$ PASS ☐ When ACH(50) is less than 3, Mechanical Ventilation installation must be verified by building department. Method for calculating building volume: ☐ Retrieved from architectural plans ☒ Code software calculated ☐ Field measured and calculated			
R402.4.1.2 Testing. The building or dwelling unit shall be tested and verified as having an air leakage rate not exceeding seven air changes per hour in Climate Zones 1 and 2, and three air changes per hour in Climate Zones 3 through 8. Dwelling units with an air leakage rate less than three air changes per hour shall be provided with whole-house mechanical ventilation in accordance with Section R403.6.1 of this code and Section M1507.3 if the <i>Florida Building Code, Residential</i>. Testing shall be conducted in accordance with ANSI/RESNET/ICC 380 and reported at a pressure of 0.2 inch w.g. (50 Pascals). Testing shall be conducted by either individuals as defined in Section 553.993(5) or (7), <i>Florida Statutes</i>, or individuals licensed as set forth in Section 489.105(3)(f), (g), or (i) or an approved third party. A written report of the results of the test shall be signed by the party conducting the test and provided to the <i>code official</i>. Testing shall be performed at any time after creation of all penetrations of the <i>building thermal envelope</i>. During testing: 1. Exterior windows and doors, fireplace and stove doors shall be closed, but not sealed, beyond the intended weatherstripping or other infiltration control measures. 2. Dampers including exhaust, intake, makeup air, back draft and flue dampers shall be closed, but not sealed beyond intended infiltration control measures. 3. Interior doors, if installed at the time of the test, shall be open. 4. Exterior doors for continuous ventilation systems and heat recovery ventilators shall be closed and sealed. 5. Heating and cooling systems, if installed at the time of the test, shall be turned off. 6. Supply and return registers, if installed at the time of the test, shall be fully open. 7. If an attic is both sealed and insulated at the roof deck, interior access doors and hatches between the conditioned space volume and the attic shall be opened during the test and the volume of the attic shall be added to the conditioned space volume for purposes of reporting the infiltration volume and calculating the air leakage of the home.			
Testing Company			
Company Name: _____ Phone: _____ I hereby verify that the above Air Leakage results are in accordance with the 2023 8th Edition Florida Building Code Energy Conservation requirements according to the compliance method selected above. Signature of Tester: _____ Date of Test: _____ Printed Name of Tester: _____ License/Certification #: _____ Issuing Authority: _____			